COER UNIVERSITY, ROORKEE
Department of Computer Science & Engineering
Course: Full Stack Web Development
LAB SHEET 4 — Node.js
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TASK 1 — CUSTOM MODULE

FILE: sum.js

```
function sum(a, b) {
return a + b;
}

module.exports = sum;
```

FILE: even-odd.js

```
function evenOdd(num) {
if (num % 2 === 0) {
return "Even";
} else {
return "Odd";
}
}

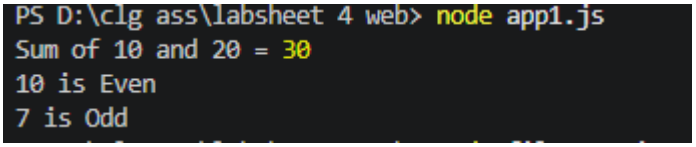
module.exports = evenOdd;
```

FILE: app1.js

```
const sum = require('./sum');
const evenOdd = require('./even-odd');

console.log("Sum of 10 and 20 =", sum(10, 20));
console.log("10 is", evenOdd(10));
console.log("7 is", evenOdd(7));
```

RUN:
node app1.js



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TASK 2 — FILE SYSTEM CRUD ON A TEXT FILE

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FILE: file-ops.js

```
const fs = require('fs');

// Create and write to student.txt
fs.writeFile('student.txt', 'Name: YOUR_NAME\nRoll Number: YOUR_ROLL_NUMBER\n',
(err) => {
if (err) {
console.log(err);
return;
}

console.log('student.txt created and data written.');

// Append course name
fs.appendFile('student.txt', 'Course: Full Stack Web Development\n', (err) => {
if (err) {
console.log(err);
return;
}

console.log('Course name appended.');

// Read the file
fs.readFile('student.txt', 'utf8', (err, data) => {
if (err) {
console.log(err);
return;
}
```

```
console.log('\nFile Content:');
console.log(data);

// Rename the file
fs.rename('student.txt', 'profile.txt', (err) => {
if (err) {
console.log(err);
return;
}

console.log('File renamed from student.txt to profile.txt.');
});
});
});
});
```

RUN:
node file-ops.js

```
PS D:\clg ass\labsheet 4 web> node file-ops.js
student.txt created and data written.
Course name appended.

File Content:
Name: Nilay Vaishnava
Roll Number: 12345
Course: Full Stack Web Development
```

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TASK 3 — DIRECTORY OPERATIONS
==========

FILE: dir-ops.js

```
const fs = require('fs');

// Create uploads directory
fs.mkdir('uploads', { recursive: true }, (err) => {
if (err) {
```

```
console.log(err);
return;
}

console.log('uploads directory created.');

// Create 3 empty files
fs.writeFile('uploads/file1.txt', '', (err) => {
if (err) console.log(err);

fs.writeFile('uploads/file2.txt', '', (err) => {
if (err) console.log(err);

fs.writeFile('uploads/file3.txt', '', (err) => {
if (err) console.log(err);

console.log('Three files created.');

// List files
fs.readdir('uploads', (err, files) => {
if (err) {
console.log(err);
return;
}

console.log('Files in uploads folder:');
console.log(files);

// Delete one file
fs.unlink('uploads/file2.txt', (err) => {
if (err) {
console.log(err);
return;
}

console.log('file2.txt deleted.');
});
});
});
});
});
});
```

RUN:
node dir-ops.js

```
PS D:\clg ass\labsheet 4 web> node dir-ops.js
uploads directory created.
Three files created.
Files in uploads folder:
[ 'file1.txt', 'file2.txt', 'file3.txt' ]
file2.txt deleted.
```

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TASK 4 — PATH MODULE

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FILE: path-demo.js

```
const path = require('path');

const filePath = '/home/user/data/report.pdf';
const relativePath = './data/report.pdf';

console.log('Directory Name:', path.dirname(filePath));
console.log('Base Name:', path.basename(filePath));
console.log('Extension:', path.extname(filePath));
console.log('Absolute Path:', path.resolve(relativePath));
```

RUN:
node path-demo.js

```
Directory Name: /home/user/data
Base Name: report.pdf
Extension: .pdf
Absolute Path: D:\clg ass\labsheet 4 web\data\report.pdf
```

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TASK 5 — OS MODULE REPORT

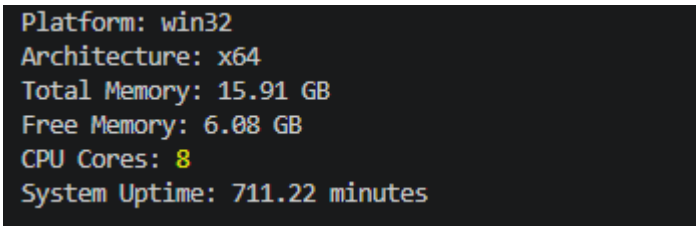
==========

FILE: sys-report.js

```
const os = require('os');

console.log('Platform:', os.platform());
console.log('Architecture:', os.arch());
console.log('Total Memory:', (os.totalmem() / (1024 ** 3)).toFixed(2), 'GB');
console.log('Free Memory:', (os.freemem() / (1024 ** 3)).toFixed(2), 'GB');
console.log('CPU Cores:', os.cpus().length);
console.log('System Uptime:', (os.uptime() / 60).toFixed(2), 'minutes');
```

RUN:
node sys-report.js



==========
TASK 6 — EVENTS MODULE / CUSTOM EVENTEMITTER
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FILE: order-events.js

```
const EventEmitter = require('events').EventEmitter;

const orderEmitter = new EventEmitter();

orderEmitter.on('orderPlaced', (order) => {
console.log('\n===== ORDER RECEIPT =====');
console.log('Order ID:', order.id);
console.log('Item:', order.item);
console.log('Amount: ₹' + order.amount);
console.log('==========================');
});

// Emit 3 orders
orderEmitter.emit('orderPlaced', {
id: 101,
item: 'Laptop',
```

```
amount: 55000
});

orderEmitter.emit('orderPlaced', {
id: 102,
item: 'Keyboard',
amount: 1500
});

orderEmitter.emit('orderPlaced', {
id: 103,
item: 'Mouse',
amount: 800
});
```

RUN:
node order-events.js

```
===== ORDER RECEIPT =====
Order ID: 101
Item: Laptop
Amount: ₹55000
========================

===== ORDER RECEIPT =====
Order ID: 102
Item: Keyboard
Amount: ₹1500
========================

===== ORDER RECEIPT =====
Order ID: 103
Item: Mouse
Amount: ₹800
========================
```

TASK 7 + TASK 8 — HTTP SERVER, MANUAL ROUTING & QUERY STRING
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FILE: server.js

```
const http = require('http');
const url = require('url');
```

```
const students = [
{ id: 1, name: 'Rahul', course: 'CSE' },
{ id: 2, name: 'Priya', course: 'IT' },
{ id: 3, name: 'Aman', course: 'ECE' }
];

const server = http.createServer((req, res) => {
const parsedUrl = url.parse(req.url, true);
const pathname = parsedUrl.pathname;

// GET /
if (req.method === 'GET' && pathname === '/') {
res.writeHead(200, { 'Content-Type': 'text/plain' });
res.end('Welcome to the Student Server!');
}

// GET /students
else if (req.method === 'GET' && pathname === '/students') {
res.writeHead(200, { 'Content-Type': 'application/json' });
res.end(JSON.stringify(students));
}

// GET /students/:id
else if (req.method === 'GET' && pathname.startsWith('/students/')) {
const id = parseInt(pathname.split('/')[2]);

const student = students.find(student => student.id === id);

if (student) {
res.writeHead(200, { 'Content-Type': 'application/json' });
res.end(JSON.stringify(student));
} else {
res.writeHead(404, { 'Content-Type': 'application/json' });
res.end(JSON.stringify({ error: 'Student not found' }));
}
}

// GET /search?keyword=node
else if (req.method === 'GET' && pathname === '/search') {
const keyword = parsedUrl.query.keyword;

res.writeHead(200, { 'Content-Type': 'text/plain' });
res.end(`Search keyword: ${keyword}`);
}
```

```
// Any other route
else {
res.writeHead(404, { 'Content-Type': 'text/html' });
res.end(`
<html>
<body>
<h1>404 - Page Not Found</h1>
</body>
</html>
`);
}
});

server.listen(3000, () => {
console.log('Server running at http://localhost:3000');
});
```

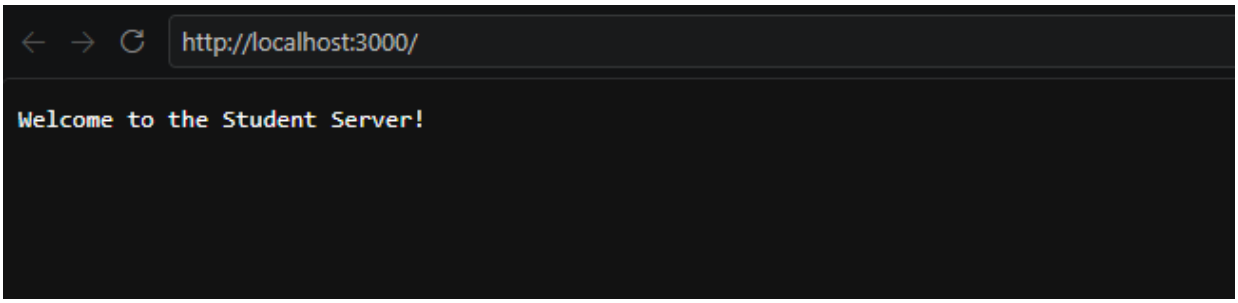
RUN:
node server.js

TERMINAL OUTPUT:
Server running at http://localhost:3000

<<< SCREENSHOT TERMINAL OUTPUT HERE >>>

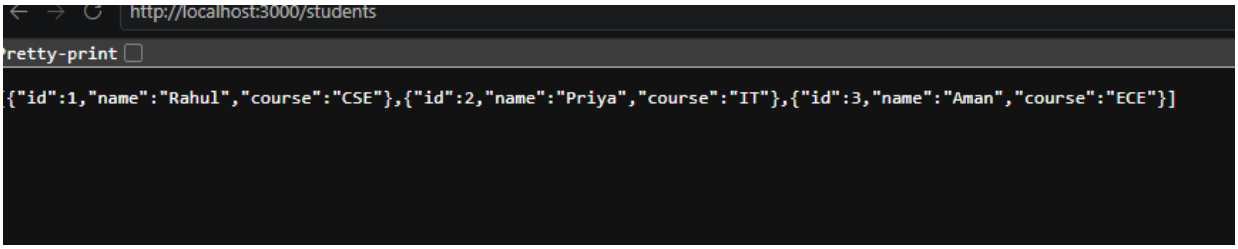
BROWSER ROUTES TO TEST:

1. GET /
URL:
http://localhost:3000/



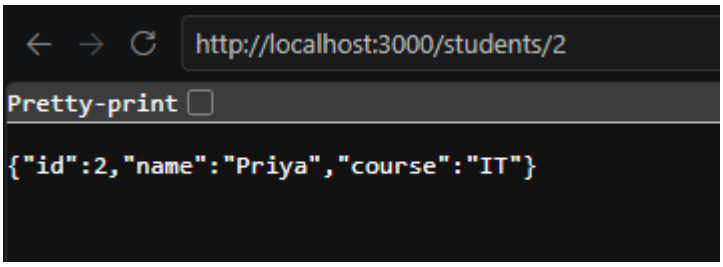
2. GET /students
URL:
http://localhost:3000/students

OUTPUT:
JSON array containing 3 students.



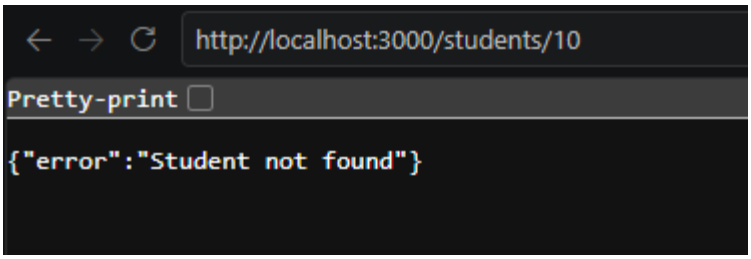
3. GET /students/:id
URL:
http://localhost:3000/students/2

OUTPUT:
{
"id": 2,
"name": "Priya",
"course": "IT"
}



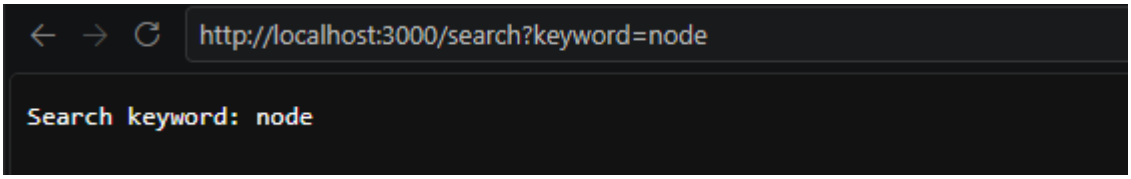
4. GET /students/:id — 404 TEST
URL:
http://localhost:3000/students/10

OUTPUT:
{
"error": "Student not found"
}



5. GET /search?keyword=node
URL:
http://localhost:3000/search?keyword=node

OUTPUT:
Search keyword: node



6. ANY OTHER ROUTE — 404 PAGE
URL:
http://localhost:3000/hello

OUTPUT:
404 - Page Not Found



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TASK 9 — READING A LARGE FILE WITH STREAMS
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FILE: stream-demo.js

```
const fs = require('fs');

// Create a file with 50 lines
let content = '';

for (let i = 1; i <= 50; i++) {
content += `This is line number ${i}\n`;
}

fs.writeFileSync('large-file.txt', content);

console.log('large-file.txt created with 50 lines.');

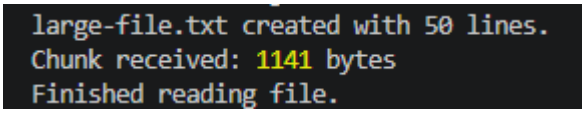
// Create read stream
const readStream = fs.createReadStream('large-file.txt');

readStream.on('data', (chunk) => {
console.log('Chunk received:', chunk.length, 'bytes');
});

readStream.on('end', () => {
console.log('Finished reading file.');
});

readStream.on('error', (err) => {
console.log('Error:', err.message);
});
```

RUN:
node stream-demo.js



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TASK 10 — npm AND EXTERNAL PACKAGE (nanoid@3)
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```
PS D:\clg ass\labsheet 4 web> npm.cmd init -y
Wrote to D:\clg ass\labsheet 4 web\package.json:

{
  "name": "labsheet-4-web",
  "version": "1.0.0",
  "description": "",
  "main": "app1.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1",
    "start": "node server.js"
  },
  "keywords": [],
  "author": "",
  "license": "ISC",
  "type": "commonjs",
  "dependencies": {
    "nanoid": "^3.3.19"
  },
  "devDependencies": {}
}


PS D:\clg ass\labsheet 4 web> npm.cmd install nanoid@3

up to date, audited 2 packages in 2s

1 package is looking for funding
  run `npm fund` for details

found 0 vulnerabilities
```

FILE: id-gen.js

```
const { nanoid } = require('nanoid');
```

```
for (let i = 1; i <= 5; i++) {
console.log(`ID ${i}:`, nanoid());
}
```

RUN:
node id-gen.js

```
  ID 1: yPl1HqcO8tqHhNFQDZPZM
  ID 2: _1RvK_E8ckrGSlrh9hMyD
  ID 3: CHHdOKrvWekMpNuTSvtHH
  ID 4: H689L70eXfQGCPJ3ptLRf
  ID 5: o2T-YbNMllGCMlRjBIdWa
```

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TASK 11 — COMMAND-LINE ARGUMENTS
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=======

FILE: calc-cli.js

```
const num1 = parseFloat(process.argv[2]);
const num2 = parseFloat(process.argv[3]);
const operator = process.argv[4];

if (isNaN(num1) || isNaN(num2)) {
console.log("Please provide two valid numbers.");
process.exit();
}

let result;

switch (operator) {
case "+":
result = num1 + num2;
break;

case "-":
result = num1 - num2;
break;

case "*":
result = num1 * num2;
break;

case "/":
```

```
if (num2 === 0) {
console.log("Cannot divide by zero.");
process.exit();
}
result = num1 / num2;
break;

default:
console.log("Invalid operator. Use +, -, *, or /.");
process.exit();
}

console.log(`${num1} ${operator} ${num2} = ${result}`);
```

RUN:
node calc-cli.js 10 5 +

```
 10 + 5 = 15
```

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TASK 12 — ENVIRONMENT VARIABLES
==========

FILE: env-demo.js

```
const http = require('http');

const PORT = process.env.PORT || 4000;

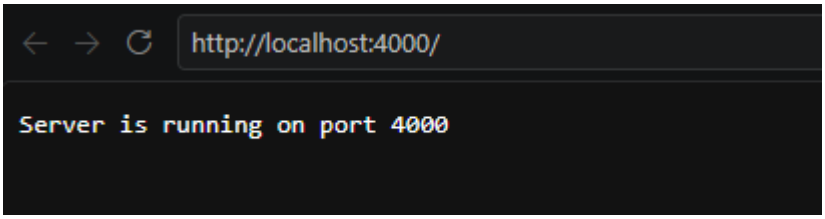
const server = http.createServer((req, res) => {
res.writeHead(200, { 'Content-Type': 'text/plain' });
res.end(`Server is running on port ${PORT}`);
});

server.listen(PORT, () => {
console.log(`Server started at http://localhost:${PORT}`);
});
```

TEST 1 — DEFAULT PORT:
node env-demo.js

EXPECTED TERMINAL OUTPUT:
Server started at http://localhost:4000

OPEN IN BROWSER:
http://localhost:4000



TEST 2 — WINDOWS POWERSHELL, PORT 5000:
$env:PORT=5000; node env-demo.js

EXPECTED TERMINAL OUTPUT:
Server started at http://localhost:5000

OPEN IN BROWSER:
http://localhost:5000

